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NEUROCRITICAL CARE SOCIETY

LESSONS LEARNED

A Case of Guillain–Barré Syndrome and Repurposing cEEG to Enable Communication in Total Locked-in Syndrome

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The Case

On presentation, a 19-year-old male patient with a history of testicular torsion reported that earlier that day, he rapidly developed numbness and tingling sequentially of his right foot, right hand, left foot, and left hand. In the emergency department, he had difficulty walking. A little more than 2 weeks prior, the patient had a sore throat, cough, and fatigue; other family members in his household also had similar symptoms. Noncontrast computed tomography scans of the head and cervical spine were unremarkable, and he was admitted to the hospital. He rapidly developed profound quadriparesis. He was intubated for failure to protect his airway, aspiration, and hypoxemia around 30 h after the first reported neurological symptoms. He developed aspiration pneumonia shortly after. He was transferred to a university-based hospital neurointensive care unit. Upon admission to the neurointensive care unit, the patient was alert and could follow commands with slight voluntary movement of his toes and right thumb. There was bilateral ptosis and decreased abduction. His limbs were flaccid, and deep tendon reflexes were absent.

What Other Information Would you Want to Obtain?

Further diagnostic steps included contrasted magnetic resonance imaging (MRI) of his neuroaxis (performed given multiple cranial nerve involvement on examination), lumbar puncture, electromyography/nerve conduction studies, and testing for antiganglioside antibodies. MRI of the brain and spine with contrast showed enhancement of multiple cranial nerves, including the left oculomotor, right abducens, bilateral hypoglossal, right spinal accessory nerves, and diffuse enhancement of the spinal nerve roots and proximal cauda equina (Fig. 1). A lumbar puncture approximately 36 h after onset was significant for a mild elevation in the protein level (47 mg/dL, normal 15–45 mg/dL) without pleocytosis (2 cells/μL). A presumptive diagnosis of Guillain–Barré syndrome (GBS) was made. Electromyogram/nerve conduction studies (EMG/NCS) on hospital day two revealed severe primarily motor polyneuropathy with some evidence of demyelination. Repeat EMG/NCS on hospital day eight was consistent with fulminant demyelinating sensory-motor neuropathy. A repeat lumbar puncture done 7 days after symptom onset was traumatic but revealed more significant albuminocytologic dissociation even upon correction for the red blood cell count (red blood cell count 2999 cells/μL, white blood cell count 20 cells/μL, protein level 374 mg/dL). These data and positive antiganglioside antibodies (anti-GM1 and anti-GD1b) corroborated the diagnosis of GBS. His paralysis continued to ascend until he was able to move only his right brow. The patient used his brow elevation to communicate answers to yes/no questions with the clinical team.

What is the Treatment for GBS?

Separate meta-analyses have shown benefit for the use of both intravenous immunoglobulin (IVIg) and plasma exchange (PLEX) in GBS [1, 2]. Data suggest that IVIg is at least as effective as PLEX and possibly that IVIg is superior [3, 4] and has a safer side-effect profile [5]. He was treated with a five-day course of IVIg. Sixteen days later, given the severity and

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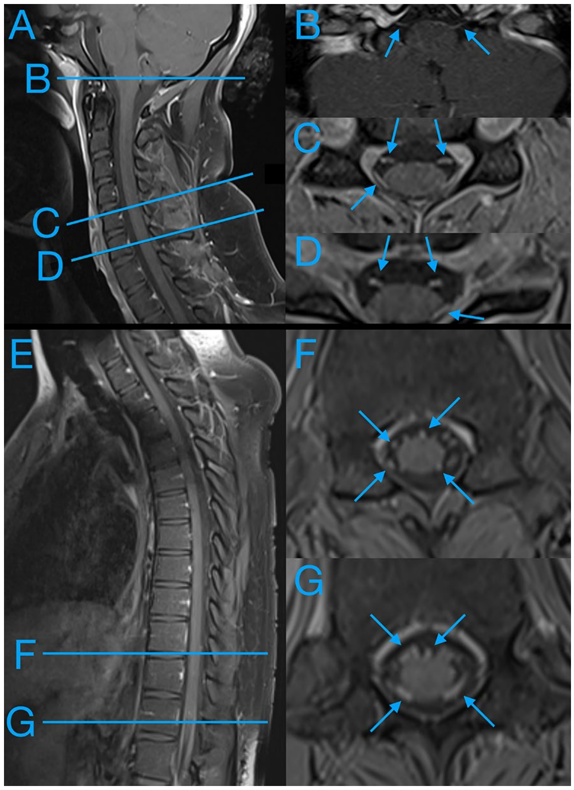

Fig. 1 Cranial nerve and cervical and thoracic nerve rootlet enhancement. Postcontrast T1-weighted magnetic resonance imaging (MRI) sequences reveal cranial nerve and cervical and thoracic nerve rootlet enhancement. **a** Sagittal cervical T1 postcontrast MRI sequence with cross-section lines corresponding to panels (**b**–**d**). **b** Cross-section at approximately the pontomedullary junction with faint bilateral cranial nerve enhancement. **c** and **d**, Cross-section at C5 (**c**) and C6 (**d**) demonstrating dorsal and ventral nerve rootlet enhancement (arrows). **e** Sagittal thoracic T1 postcontrast MRI sequence with cross-section lines corresponding to panels (**f**, **g**). **f** and **g**, Cross-section at T5 (**f**) and T11 (**g**) demonstrating dorsal and ventral nerve rootlet enhancement (arrows)

progression of his GBS, he was also treated with PLEX. However, combing PLEX with IVIg has not shown to provide further benefit [4].

Steroids are not an effective option in GBS, as they do not change disability after four weeks, nor do they affect rates of death or disability at 1 year [6]. Despite evidence that suggest steroids may be effective in experimental animal models of autoimmune neuritis, there is some relatively low-quality evidence in humans to suggest corticosteroids may actually delay recovery [6]. It has been proposed that the benefits of steroid-based immunosuppression are counteracted by steroid inhibition of repair processes [6].

Case Continued

Despite early treatment with a five-day course of IVIg (2 g/kg total dose divided across five days) the patient's condition worsened, leading to complete loss of voluntary movement by hospital day five. Continuous electroencephalography (cEEG) surface electrodes were placed over the forehead to amplify nonvisible facial muscle activity (right brow muscles targeted because they were the last to lose all visible movement). This facilitated communication by allowing staff and family members to visualize muscle activity in real-time on a cEEG screen. This electrophysiologic communication interface allowed the patient to communicate his level of pain, anxiety, and other concerns despite being in a total locked-in syndrome [7]. Despite aggressive treatment, he remained quadriplegic and dependent on mechanical ventilation by hospital day 40, with only partial recovery of some cranial nerve function.

Use of cEEG for Communication

The patient was no longer able to communicate when the last observable voluntary brow movement was lost. An electrophysiologic communication interface was created using cEEG surface electrodes to amplify nonvisible muscle activity (see Supplementary Material for specifications). Real-time muscle amplification peaks were easily detected by medical staff and family members on a live cEEG monitor. With minimal training, they used a code in which two peaks indicated "yes" and one peak indicated "no," enabling the patient to communicate his level of pain, anxiety, and shortness of breath, as well as answer yes/no questions posed by the clinical team (Fig. 2; Supplementary Video; see Supplementary Material for detailed method to implement this approach).

Family members noted it was "super easy" to learn to use the electrophysiologic communication interface. They were also appreciative of regaining some ability for the patient to communicate, stating that "even though there were no signs of life and he was not breathing on his own, the fact that he could hear us and respond meant he is alive." This simple yet effective electrophysiologic communication interface provided a practical approach to addressing communication challenges in severely paralyzed individuals, thereby enhancing patient autonomy and improving the quality of care. The patient eventually regained enough function to nod and shake his head, make some eye movements, open and close his mouth, and shrug his shoulders prior to being discharged to a ventilator-capable long-term acute care facility. The patient was discharged from the facility; he is currently ventilator-liberated and wheelchair-bound and, with hand orthoses, is able to use a computer and feed himself.

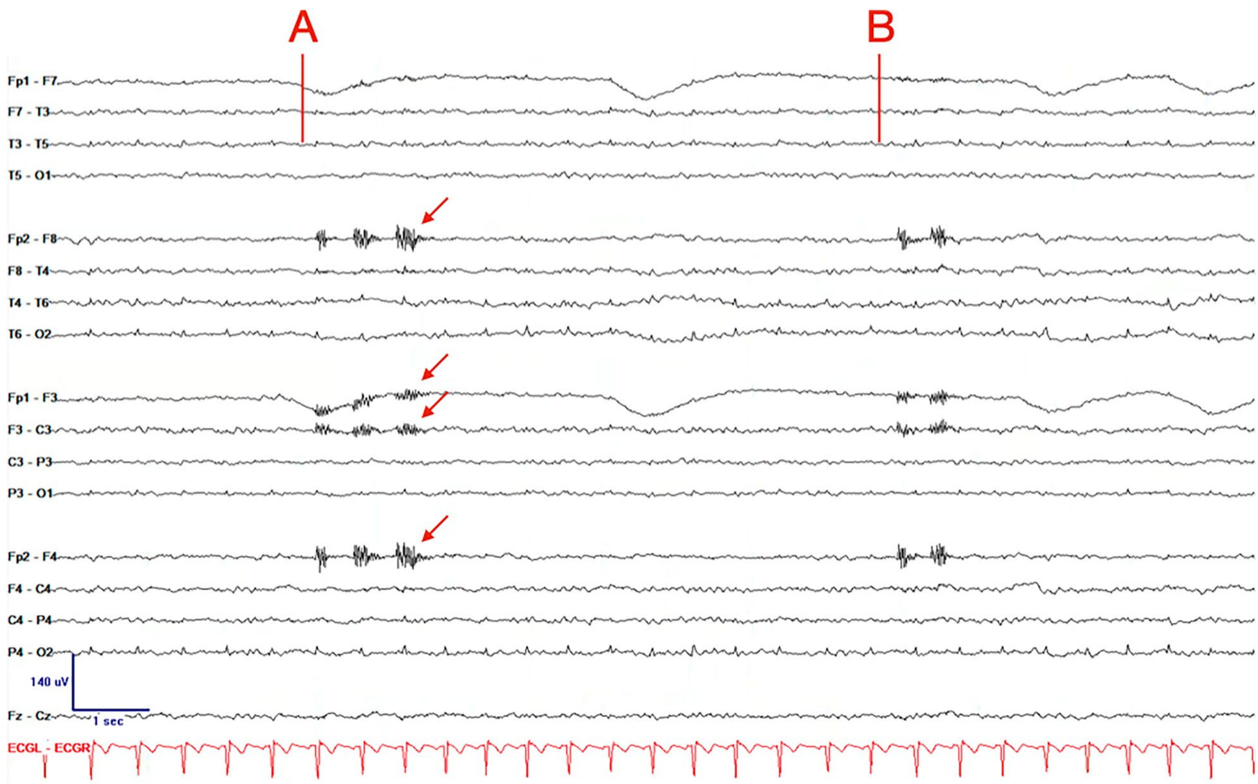

Fig. 2 EEG enables communication by amplifying muscle artifacts. Screenshot from EEG tracing demonstrating forehead muscle signal amplification, enabling communication in a patient with total locked-in syndrome. **a**, **b** The patient was asked to raise their eyebrows three (**a**) and two times (**b**). Discrete amplified frontalis muscle signal is observed in the frontal EEG leads (Fp2-F8, Fp1-F3, F3-C3, Fp2-F4). Arrows highlight example amplification artifacts. AP bipolar montage shown; vertical and horizontal scale bars depict 140 µV and 1 s, respectively. A 60-Hz notch filter, a 1-Hz low-frequency filter, and a 70-Hz high-frequency filter were used. Sensitivity was set to 70 µV/mm, and a time base of 30 mm/s was used. AP, anteroposterior; EEG, electroencephalography

Discussion

Here, we present a compelling case of a 19-year-old man who presented to the emergency department with symptoms of GBS that were preceded by an upper respiratory tract infection. Despite initial treatment with IVIg and PLEX, the patient developed profound quadriparesis and respiratory failure. Notably, the patient lost all voluntary movement, becoming totally locked-in. cEEG was used in a nontraditional way as an electrophysiologic communication interface to amplify forehead muscle electrographic signals to enable basic communication. Medical staff and family members were easily able to detect real-time amplification peaks. This allowed them to rapidly begin asking the patient questions and deciphering answers.

Many complex and more invasive ways have been proposed to facilitate communication in locked-in patients [8, 9]. The approach is remarkably simple: using surface EEG electrodes to amplify nonvisible muscle activity. In this case, we targeted the brow region; however, a simple approach could be used for alternative face or body muscles given different patient circumstances (i.e., targeting muscles the patient last lost control of). This electrophysiologic communication interface enabled the patient to communicate his level of pain, anxiety, and shortness of breath, as well as answer yes/no questions posed by the clinical team. It also helped rule out status epilepticus contributing to his lack of responsiveness.

Use of cEEG in this context represents a novel approach to improve the communication challenges faced by severely paralyzed individuals [9]. By amplifying nonvisible muscle activity, this simple electrophysiologic communication interface enabled a patient to communicate effectively with the clinical team, despite being in a total locked-in syndrome. This straight-forward approach uses a technology accessible to most intensive care units and can relatively easily enhance patient autonomy and hospitalized quality of life.

Lessons Learned

1. *Rapid progression and early diagnosis* GBS can quickly escalate from mild to severe symptoms, emphasizing the necessity of early recognition and comprehensive diagnostic steps, including MRI, lumbar puncture, and nerve conduction studies.
2. *Effective treatment protocols* IVIg and PLEX are the primary treatments for GBS, with IVIg often preferred because of its safety profile. Steroids are not effective and may delay recovery.
3. *Communication in severe cases* For patients who progressed to a total locked-in syndrome due to severe GBS, cEEG can be used as an electrophysiologic communication interface to amplify nonvisible muscle activity, enabling basic communication.
4. *Accessibility and adaptability of cEEG* cEEG is widely available in intensive care units. Our approach can be implemented without additional software, providing a practical and adaptable solution to improve patient autonomy and quality of care.

Supplementary Information

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Author contributions

Patient care: JML. Conceptualization: TJP, JML. Writing–original draft: JML. Writing—review and editing: TJP, JML. Generative AI was used for enhancing manuscript clarity and improving grammar.

Source of Support

None to report.

Conflicts of Interest

There are no potential conflicts of interest relevant to this article.

Ethical Approval/Informed Consent

This study is in accordance with the principles of the Institutional Review Board (IRB) of the University of Pennsylvania that a case report of three or less cases does not require IRB approval (3.2.2). The patient consented for their case to be published.

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